MorletDB: Recordings, Not Rows — Lag-Tolerant Waveform Retrieval in PostgreSQL

Version 0.17

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ABSTRACT

Analysts of waveform archives search for intervals that resemble a query up to a small time shift, and they need the original samples of every hit. We treat this as top-k search under a bounded-lag cross-correlation scorer inside PostgreSQL. First, we show why standard vector search fails at it: on 235,000 seismic windows, a zero-lag shortlist of 1000 recovered only 74% of the scorer's top ten, and an exact shortlist lost the same matches as an HNSW one. The candidate metric, not approximate search, loses them. Second, we make the scorer itself searchable. Each lag term is an inner product with a zero-padded, shifted copy of the query, so 13 shifted probes against one unmodified inner-product HNSW index, re-ranked with the unchanged scorer, generate candidates for the scorer actually used, with no training, no per-shift copies of the data and no lower bounds. On held-out stations and year, probe search recovered 97.7% of the exact top ten at a p95 latency of 57 ms, against 740 ms for in-database exact search; zero-lag candidates from the same index recovered 0.639–0.658. Exact candidates on the same sparse grid recovered 99.9%. Against in-database exact search made 1.6–2.7× faster by a one-pass scorer and 3.2–3.7× faster again by a float32 screen with a rigorous error bound, both with identical results, probes stayed 1.6–2.3× faster at p95 on galleries of up to one million windows. On a proxy task, the scorer returned windows containing catalogued earthquakes more often than zero-lag similarity, on development data (precision at ten 0.297 against 0.213) and on held-out stations and year (0.139 against 0.060). Third, MorletDB keeps one search contract on both paths: every hit is a sample interval that reads back the losslessly stored recording bit for bit, and filters and row-level security apply to exact and probe search alike. Probe structures cost 0.97–1.34× the raw data. Exact search remains the choice for small or selective galleries, and the router sends those to it. An exact scan of a materialized copy outside the database was faster still, without the contract's filters, security and commit-time visibility.

1 INTRODUCTION

Condition monitoring, seismology and physiology produce archives of long, uniformly sampled recordings; condition monitoring of machines motivated this work. A common investigation starts from one interval and asks where similar behaviour occurred, then inspects the original samples and acquisition context of each candidate. Two practical requirements follow. Similarity must tolerate a small time shift, because a window grid rarely aligns two occurrences of the same event; and a result is useful only if it identifies an interval of a recording that can be read back exactly.

A bounded-lag cross-correlation captures the first requirement: two windows are similar if some shift of at most L samples correlates them strongly (Section 2). Exhaustive evaluation is linear in the archive. The obvious acceleration is a vector index, which PostgreSQL users already have through pgvector [1]; but such indexes rank by a fixed metric between unshifted vectors. This paper asks:

> *How can bounded-lag waveform search be accelerated with standard vector indexes while preserving the link to the recording, the semantics of filters and transactional updates, and which costs limit its use?*

The answer has three parts. Zero-lag vector search asks the index the wrong question: it ranks by a different quantity and loses a quarter of the scorer's top ten, and an exact zero-lag shortlist loses the same windows as an approximate one (Section 6). The right question is a set of shifted inner-product queries: each lag term of the scorer is an inner product with a shifted query, so a few shifted probes against one index generate candidates for the scorer actually used (Section 3). And PostgreSQL can answer it without giving up what makes a hit useful: candidates are re-ranked with the scorer and returned as sample intervals of losslessly stored recordings, under the same filters and access rules as exact search (Section 4).

Contributions.

(1) **A diagnosis.** Zero-lag shortlists fail for a bounded-lag scorer because of their metric, not because of approximate search: exact and HNSW shortlists lose the same matches (Section 6).
(2) **A simple method.** One unmodified inner-product HNSW index serves a bounded-lag scorer: query-side, zero-padded shifted probes on a sparse lag grid, the union of their candidates, and re-ranking with the unchanged scorer. It needs no trained embedding, no per-shift copies of the data and no lower bounds (Section 3).
(3) **One contract on both paths.** Exact and probe search share eligibility, metadata filters, row-level security, source exclusion and overlap suppression. Every hit is an interval that reads back the original bits, and windows become searchable at the commit that builds them. A router sends selective filters to exact search (Section 4).
(4) **A controlled evaluation.** Protocols frozen before measurement, held-out stations and year, an ablation that separates grid sparsity, float16 probes and index error, and measured costs of filters, storage and index maintenance (Sections 6–9).

Findings in brief. (1) *What works:* on held-out data, shifted probes recovered 97.7% of the bounded-lag top ten at a p95 latency of 57 ms, against 740 ms for in-database exact search (Table 2), and 1.6–2.3 times faster at p95 than an in-database exact path that, with identical results, screens windows under a rigorous error bound and re-scores only possible neighbours. (2) *Why:* zero-lag candidates miss most matches whose best lag exceeds ±5 values, exactly or with HNSW (Figure 2); probes on a sparse grid cover all lags

(Figure 3, Table 3). (3) *Cost:* probe structures of about the size of the raw data, 13 index passes per query, routing for selective filters, and write-ahead logging from index maintenance (Sections 8 and 9). (4) *Relevance proxy:* the bounded-lag top ten contained catalogued earthquakes more often than the zero-lag top ten, by 0.08 in precision at ten on development and on held-out data (Section 11). All retrieval results are for seismic recordings; Section 11 states the limits.

Code, frozen protocols and raw results are public [5]; a supplement reports storage, recovery and further studies.

2 PROBLEM AND SEARCH CONTRACT

A *recording* is a sequence $x_0, \ldots, x_{N-1}$ of float64 samples with a sample rate, a channel, a unit and JSON acquisition metadata. An *interval* $[a, b)$ denotes $x_a, \ldots, x_{b-1}$; sample offsets, not wall-clock times, are authoritative. A *window* is an interval of fixed length w on a stride grid. A search representation $\phi(x_{a:b}) \in \mathbb{R}^n$ is derived from a window by a versioned extractor.

We study envelopes: for $w = 4096$, z-score the window, take the magnitude of the analytic signal, smooth with a 50-sample moving mean, keep every tenth value and z-score again ($n = 410$). Stored envelopes are float16. For a query envelope q computed at full precision and a stored envelope g, the scorer is

$$D(q, g) \;=\; 1 - \max_{|\ell| \le L} \; \frac{1}{n} \sum_{t} q_{t+\ell}\, g_t, \qquad L = \lfloor 0.075\, n \rfloor = 30, \quad (1)$$

where terms with $t + \ell$ outside $[0, n)$ are zero. This is a *biased* cross-correlation: no per-lag renormalization of the overlapping part, no circular shift, and one z-score per window rather than per overlapping subsequence. At 100 Hz, $L = 30$ envelope values correspond to ±3 s.

Contract. A query names a source interval (or supplies samples), a metadata filter and k. The *eligible* windows are those whose recordings have the query's sample rate, channel and unit, whose window size and extractor match, whose metadata contain the filter, and which the caller may see under row-level security. The *exact result* is the first k eligible windows in the order $(D, \text{recording}, \text{start})$, after excluding windows overlapping the source interval and greedily skipping windows that overlap a better window of the same recording. Each result row is a recording identifier, an interval and its distance; the interval reads back the original samples bit-exactly from the raw store.

Three kinds of exactness. We keep apart (i) *exact distances*: every returned distance is D computed on the stored envelope; (ii) *exact candidate selection*: the returned set equals the exact result; and (iii) *relevance*: the returned intervals help answer the analyst's question. Probe search guarantees (i) but not (ii); we measure (ii) as recall@10 against the exhaustive search; neither implies (iii).

3 SHIFTED INNER-PRODUCT PROBES

For a lag ℓ, define the zero-padded shifted query $s_\ell[t] = q_{t+\ell}$ if $0 \le t + \ell < n$ and 0 otherwise. Then $\sum_t q_{t+\ell} g_t = \langle s_\ell, g \rangle$ and

$$1 - D(q, g) \;=\; \frac{1}{n} \max_{|\ell| \le L} \langle s_\ell, g \rangle. \quad (2)$$

The bounded-lag top-k is therefore the top-k of a maximum over $2L{+}1$ inner-product queries against the *unshifted* stored vectors.

Algorithm 1 Probe search (product function, simplified)

Require: query envelope q (full precision), eligibility predicate E, k; grid $\Lambda = \{-L, -L{+}5, \ldots, L\}$; per-probe budget $c = 50$; threshold $T = 5000$

```
if |{g : E(g)}| ≤ T then ▷ counted from recordings, stopping at T
    return ExactScan(q, E, k)
end if
C ← ∅
for ℓ ∈ Λ do
    C ← C ∪ top-c by ⟨f16(s_ℓ), g⟩ over g with E(g)    ▷ HNSW, iterative scan
end for
score every g ∈ C with D(q, g) on its stored envelope
return first k in order (D, rec, start) after source exclusion and overlap suppression
```

Two consequences matter. The shift is applied to the query, so the index holds one copy of each window, not one per lag. And the probes must be inner-product queries, not cosine: $\|s_\ell\|$ shrinks with $|\ell|$ because of zero padding, exactly as the biased scorer weights a shifted overlap.

Re-ranking uses the full-precision query and the stored envelope, so every returned distance is exact; only the set of windows considered is approximate. Algorithm 1 replaces the full lag set by a sparse grid ($|\Lambda| = 13$ at spacing 5, chosen on development data) and each exact inner-product top-c by an HNSW search. Three approximations follow, and Section 7 measures each separately: (a) *grid sparsity*: a window whose best lag lies between grid points must rank in the top c of a neighbouring probe, which smooth envelopes favour; (b) *per-probe budget*: a global top-ten window need not be in any single probe's top c; (c) *index approximation*: HNSW returns an approximate top-c and compares a float16 copy of the probe with float16 stored vectors. The budget is fixed: if source exclusion or overlap suppression leaves fewer than k windows, the function returns fewer rows, where the exact path fetches more. This never happened in our evaluation.

4 INTEGRATION IN POSTGRESQL

MorletDB is a PostgreSQL 17 extension written in Rust with pgrx. Figure 1 shows the paths that matter for search; the supplement describes storage.

Recordings, not rows. The user-facing unit is a recording, and every search result points into one. Domain sequence types have precedent, for example the temporal types of MobilityDB and MEOS [36, 37]; our recordings use integer sample offsets and half-open intervals rather than interpolated continuous time, and their coding is bit-exact. The scorer and the envelope exist once, in a Rust core used by the extension; evaluation code re-implements them in Python, checked against the core through fixtures.

Name. The system is named in honour of Jean Morlet, after its starting hypothesis that Morlet-wavelet representations suit search in technical signal archives. Representations are versioned, replaceable extractors; the envelope used here is not a wavelet.

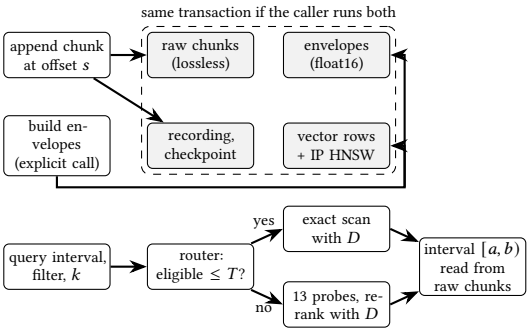


Figure 1: Write path and the two search paths. An append writes the raw chunk and advances the checkpoint; envelopes and, when probe search is enabled, their vector rows are written by an explicit builder call. All of them commit together when the caller runs append and builder in one transaction. Both search paths score candidates with D on stored envelopes and return sample intervals, which are read from the raw chunks, not from a representation. The exact path reads envelopes; the probe path reads vector rows for candidates and envelopes for re-ranking.

Recordings and appends. A collection is a set of caller-owned tables: recordings with a checkpoint, raw chunks, envelopes and, optionally, a pgvector table of `halfvec` rows with an inner-product HNSW index. An append names its global start offset and writes the raw chunk and the checkpoint in one transaction, so a committed checkpoint never refers to uncommitted samples. A repeated identical append is a verified no-op; a gap, overlap or changed retry fails. Raw chunks are coded losslessly, so a result interval reads back the appended float64 bit patterns. The codec rests on one observation: most sensor floats are integers in disguise, an affine image $o + k\,s$ of digitizer counts under floating-point rounding. Given the recording's nominal quantum, or a grid it detects per chunk when none is declared, the encoder recovers the integer indices, codes them with polynomial prediction and Rice codes, and stores last-place corrections where rounding departs from the grid. On calibrated physical values (Paderborn bearing data, Sleep-EDF EEG) it was 1.5–1.6× smaller than zstd, where the tested ALP implementation fell back to near-raw size, and smaller than ClickHouse 26.5 and TileDB with tuned codec chains (0.62–0.76×); on counts it was within 5% of FLAC (supplement). Against the 2026 float codecs it remained the smallest on four of six waveform inputs, but Pcodec was 5–8% smaller on two inputs that lie almost, not exactly, on a grid (IMS decimals and the Paderborn physical values), as was ClickHouse 26.10's PCO codec by 3% on the latter; every ALP arm was 2.6–7.9× larger on calibrated values [45]. Detection without a declared quantum matched the declared-quantum size on all six inputs tested (ratio 0.98–1.00), so the advantage does not depend on the user supplying calibration metadata [49]. Against QuestDB 10.1 on 101.6 M samples of Paderborn bearing data, MorletDB stored the sample data in 33% less space than QuestDB's Parquet zstd 9 arm and 87% less than its native columns; it read 4,096-sample intervals and computed interval statistics faster, tied on long reads, ingested 1.2–1.6× more slowly from one client connection and 1.2–2.0× faster with parallel writers. On data without a value grid it would lose most of the storage advantage (supplement [44]). *An append does not build search representations.* Envelopes are written by an explicit call to `morlet_build_envelopes`, which also writes their vector rows if probe search was enabled for that window size when the call started. Samples and their representations become visible together only when the caller runs the append and the builder in the same transaction:

```
BEGIN;
SELECT morlet_append_chunk('signals', 'pump7', 81920, :x);
SELECT morlet_build_envelopes('signals', 'pump7');
COMMIT;   -- chunk, checkpoint, envelopes, vector rows
```

A builder call writes at most `max_windows` windows (10,000 by default) and returns how many it wrote; a call that hits this bound leaves later complete windows unbuilt until the next call, so a caller that needs every window covered repeats it until it returns 0. Windows of a representation may cross chunk boundaries, so a window becomes buildable only when the append that completes it has been made. Envelopes and vector rows are derived and can be dropped and rebuilt without touching raw data.

Summaries, acquisitions and small machines. Optionally, each chunk also stores a 66-byte summary (count, minimum, maximum and compensated sums). Interval statistics and overviews then decode only the two edge chunks, with results bitwise identical to decoding every chunk: 58× faster at p95 for 60 s of 64 kHz vibration, and an hour of it (230 M samples) in 36 ms, for 4.5% slower ingestion and 1.3% more chunk storage [46]. Acquisitions group channels recorded together; one call reads all of them over a time interval, 32–39% faster at p50 than QuestDB for the seven channels of a B3 acquisition. Computing search features during ingestion cost 3.9× the ingest time of raw storage on the B3 data (1.64 against 6.33 M samples/s from one connection) [44]. The full workflow (ingestion with features and summaries, envelope build, bit-exact read-back of every sample, statistics and search) ran inside container limits of 2 CPUs and 512 MB at the speed it reached without limits [47], and on a Raspberry Pi 3B+ (1 GB, microSD card, no fan) at 78,000 samples/s with features and 258,000 without, returning the same search results as on the desktop [48].

Two search paths, one contract. Exact and probe search apply the same eligibility, exclusion, ordering and suppression, and both return distances computed by the same Rust scorer; a caller can switch paths without changing what a result means. The exact path scores every eligible stored envelope with D in one parallel SQL statement; the probe path implements Algorithm 1. All functions run as `SECURITY INVOKER`, so privileges and row-level security apply to every table read. The probe path finds windows through the probe table and the exact path through the envelope table, so they search the same windows only when every stored envelope has its vector row. `morlet_build_envelopes` writes envelopes and, when probe search is enabled, their vector rows in one call and continues after the last stored window, which keeps each recording's searchable windows a gap-free prefix; envelopes written by other processes, or out of order, can leave windows without vector rows (reported as `probe_backlog` by `morlet_collection_status` and

filled with one INSERT from the envelope table) or gaps that the status does not show. The exact functions run in the snapshot of the calling statement. The probe function issues several statements; under READ COMMITTED each takes its own snapshot, so a window appended or deleted during the call may be a candidate in one statement and absent from another, while under REPEATABLE READ or SERIALIZABLE all share one. In every case each returned row was visible, eligible and scored in the re-ranking statement's snapshot. Online index rebuilds (`REINDEX CONCURRENTLY`) keep the probe path usable while they run.

Filters. An approximate index answers "nearest overall"; a filter asks "nearest among eligible". Applying eligibility after probing silently discards candidates when few windows qualify. The router therefore counts eligible windows from the recordings table, stopping at the threshold T, and sends small galleries to the exact path; above T, each probe scans only eligible recordings, using pgvector's iterative index scan. Choosing between brute force and an index by filter selectivity is established practice in vector databases (Section 10); we include it because it decides whether probe search keeps the contract under filters.

Access control. Row-level security on the recordings table alone protects the search functions but not direct reads of the derived tables. `morlet_enable_row_security` therefore attaches a policy to every companion table that admits rows whose recording is visible; written as membership in an array of visible identifiers it keeps parallel scans (exact search +5%), whereas a correlated `EXISTS` tripled exact search time. The acquisition tables, added later, first checked a channel's acquisition only by foreign key: in SQL 0.3.1 and 0.3.2, a role that could add acquisition channels could attach a recording it sees to an acquisition it does not see and then read that acquisition's metadata. SQL 0.3.3 (release 0.3.4-alpha.1) admits channel rows only into visible acquisitions, through the function and through direct writes, with a two-tenant regression test (advisory GHSA-pg6f-hcvw-vw34).

Agent access. A read-only Model Context Protocol server exposes the archive to AI agents through twelve tools: listing collections and recordings, coverage and trends over time, locating a UTC time, describing, comparing and reading intervals, finding onsets and envelope-spectrum lines, and similarity search from an interval or a time [50]. Every call runs in a read-only transaction with a statement timeout, as the configured database role, so privileges and row-level security decide what an agent sees, and every hit it cites reads back the stored samples. We report it as a feature with case studies, not as an evaluation (supplement).

5 EVALUATION SETUP

Kinds of studies. *Confirmatory* studies committed their questions, data, arms, metrics and decision rules to the repository before their first result and ran once; deviations are logged in the protocols [5]. These are R1b, R2, BL1 and its confirmation BL1-C, BL1-S, SS1, SS3, E1, E3, E4, E5 and its held-out replication E5-H, E6, E7, B2, B3 (with its amendments B3.1–B3.3), B4, G1, CS1, EX1, EX2 and the load runs; their protocols are internal plans, not registrations with a third party. *Exploratory diagnostics* had a protocol fixed before running but no confirmatory decision attached (R2-D, D1, D2) or none at all (the pruning feasibility study). *Post hoc* are the grid-spacing explanation and the choice of illustrative cases; the selection rule for the cases was committed before the figure was made, after the results were known. The supplement lists each study with its commits. CS1, EX1 and EX2 are labelled S1, X1 and X2 in the repository; we rename them because S1 and X1 name earlier studies here. Table 1 lists the studies used here.

Data. Development data (*dev*) are CX HHZ recordings [4] of stations PB01, PB02, PB06 and PSGCX, 1–14 March 2025: 481 M samples at 100 Hz, 234,762 windows of 4096 samples at stride 2048. Held-out data (*held-out*) are stations PB05, PB11, PB14 and PB18, 1–14 March 2021 (234,622 windows), chosen from the data centre's availability listing before any waveform was downloaded; no earlier study had read them. Each set has 100 queries from its first day: 50 random and 50 high-energy windows. Seismic envelopes are one signal domain; we discuss generality in Section 11.

Consolidated comparison. The main retrieval tables come from E1, a comparison of all arms on one build. E1 re-measured the held-out data that BL1-C had already used for its single confirmation (and SS3 reused), with configurations frozen before BL1-C; it is therefore a consolidated re-measurement, not another independent confirmation. It ran pg_morlet SQL version 0.3.1 at commit 68594d7 on `main`, before that version was released; release 0.3.1-alpha.1 ships it together with the later fixes named at the end of this paragraph. E1 used one configuration, the same queries and one session, under a protocol frozen before the run [31]. PostgreSQL 17.11 ran with default settings (128 MB shared buffers, 4 MB `work_mem`, two parallel workers per query; one exact arm allowed eight, of which the planner used three; PostgreSQL 17 has no asynchronous I/O), pgvector 0.8.0 (HNSW m=16, `ef_construction` 64), warm caches, one client, on an Apple M5 Max (18 cores, 128 GB) with no other CPU-intensive job (load average 3.8–4.7, mostly E1 itself). Each arm ran twice per query; latencies use all 200 timings. In both databases every envelope had its vector row, every recording's windows formed a gap-free prefix, and nothing was written during the runs, so all arms searched the same complete set of windows. Later fixes to the extension (row-security set-up, a race between routing and concurrent commits, a race between enabling probes and concurrent builds) change none of the statements measured here [31].

Earlier studies. The studies that explain design decisions (R2-D, BL1, SS1, SS3) used earlier implementation states and a machine shared with an unrelated CPU-intensive job; we label their numbers as such. Recall@10 is always against the exhaustive exact result.

6 WHY ZERO-LAG SHORTLISTS FAIL

A zero-lag shortlist loses a quarter of the bounded-lag top ten, and approximate search is not the reason. The natural first stage takes the top K windows by zero-lag cosine from an ANN index and reranks them with D. A scaling study measured this on dev galleries nested from 4,217 to 234,762 windows [7]. Its statement passed the query vector through a subquery, which pgvector cannot use as an index ordering key, so it computed an *exact* zero-lag top K. A diagnostic on the same data, with its protocol frozen before measuring, then compared an explicitly exact shortlist with an HNSW shortlist whose plan was checked to use the index.

Table 1: Studies, data and measurement kind.

Study	Question	Data
R2, R2-D	exact scaling; zero-lag shortlists, exact vs HNSW	dev
BL1	probe grids and budgets, ablation controls	dev
BL1-C	frozen configuration, once	held-out
SS3	filtered search and router; product check	dev, held-out
SS1, R1b	one stored copy; float16 precision	dev, held-out; 2022
E1	**all arms on one build**	**dev, held-out**
E3, E4	materialized exact scan outside the database; 0.23–1.03 M windows	dev, days 1–61
E6, E7	faster in-database exact scorer; screened exact search; identical results	dev, days 1–61
E5, E5-H	catalogued earthquakes, bounded vs zero lag	dev; held-out
B3, G1	storage, ingest, reads vs QuestDB; grid detection	Paderborn; B1 inputs
B4	codec vs 2026 float codecs (Pcodec, ALP, ClickHouse PCO)	B1 inputs; Elf series
CS1	interval statistics from per-chunk summaries	Paderborn; dev
EX1, EX2	full workflow at 2 CPUs, 512 MB; on a Raspberry Pi 3B+	Paderborn
D1, D2	time per stage; filter correlation	dev
BL1-S, C2–C4	index maintenance during ingestion	dev; dev+held-out

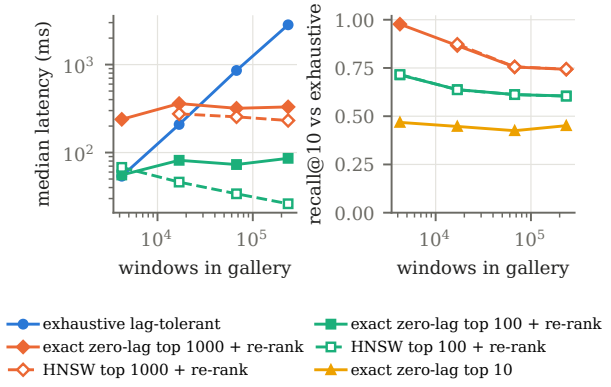


Figure 2: R2, R2-D on dev galleries. Left: median latency of the exhaustive scan (historic, single-threaded implementation) and of zero-lag shortlists plus re-ranking (exact: solid; HNSW: dashed). Right: recall@10; the HNSW curve lies on the exact one. 100 queries per gallery.

With K=1000, recall@10 fell from 0.977 on the smallest gallery to 0.743 on the largest; with K=100, from 0.714 to 0.605 (Figure 2). The HNSW shortlist agreed with the exact one on 98–100% of candidates on the largest gallery and gave the same recall to within

Table 2: E1, held-out data (234,622 windows, 100 queries, 200 timings per arm). Recall@10: mean / minimum / queries below 0.9.

Arm	Recall@10	p50 / p95 ms	Work per query
exact, 2 workers	1	958 / 971	score all windows
exact, 8 allowed (3 used)	1	728 / 740	score all windows
probes (product)	**0.977 / 0.7 / 5**	39 / **57**	13 scans, re-rank ≈350
zero-lag HNSW, 650	0.639 / 0.1 / 69	28 / 40	1 scan, re-rank 650
zero-lag HNSW, 1000	0.658 / 0.1 / 67	34 / 50	1 scan, re-rank 1000

0.007 at every size (0.744 at K=1000). Every top-ten item contained in a shortlist survived re-ranking and suppression, so coverage equalled recall. The zero-lag candidate metric causes the loss, not approximate search and not the contract's exclusion rules: a quarter of the bounded-lag top ten lies outside the 1000 windows nearest by zero-lag cosine. The exact zero-lag top ten alone shared 43–47% of the bounded-lag top ten. E1 reproduced this on the current build: a zero-lag HNSW shortlist of 1000 from the probe index recovered 0.744 on dev and 0.658 on held-out data. The exhaustive scan in Figure 2 is the historic implementation (linear, 12.1–12.8 µs per window); the current exact path runs as one parallel statement and is the baseline below.

7 SHIFTED PROBES: ABLATION AND CONFIRMATION

The configuration was chosen on dev data [10]: grids of 7 and 13 probes and budgets of 50 and 100 per probe all met its gate (recall@10 ≥ 0.95, p95 at most a third of the exhaustive scan and below the fastest exact path, added storage at most 1.5× raw); 13 probes × 50 was frozen and confirmed once on held-out data. E1 then measured the product functions and all controls on one build [31]. Its offline controls use the same stored float16 envelopes and the same re-ranking: *PX-61*, the exact top 50 for each of the 61 lags; *PX-13* and *PX-7*, the same for the 13- and 7-lag grids; *PQ-13*, as PX-13 with each probe rounded to float16, as the index sees it; and *EZ*, an exact zero-lag shortlist with as many candidates as PX-13.

Benefit. On held-out data probe search kept 97.7% of the exact top ten at a p95 latency of 57 ms, against 740 ms for in-database exact search on the measured build (Table 2); on dev data, 99.8% at 53 against 783 ms. Zero-lag candidates from the same index recovered 64–66%, even with a larger re-rank budget. By query group, held-out recall was 0.998 for the 50 random windows and 0.956 for the 50 high-energy windows; energy is a selection rule for queries, not a measure of relevance. Of the 100 held-out queries, 83/12/4/1 reached 1.0/0.9/0.8/0.7. The 23 missed intervals (of 1,000) ranged from rank 1 to 10 in the exact list and 3 were the exact best match; lower-ranked admissible intervals took their place. Both rounds on the same index returned identical lists; a rebuilt HNSW index can return different candidates.

Against a faster exact path. On the measured build, 97% of exact search time was spent in the per-row scorer. We rewrote it to build the query's shifted copies once per query and to score each

Table 3: E1: recall@10 by candidate generator. All arms re-rank with D.

Candidates	dev	held-out
all 61 lags, exact (PX-61)	1.000	1.000
13-lag grid, exact (PX-13)	1.000	0.999
13-lag grid, float16 probes, exact (PQ-13)	1.000	0.999
7-lag grid, exact (PX-7)	0.986	0.987
13-lag grid, HNSW (product)	**0.998**	**0.977**
zero-lag, exact, same count (EZ)	0.681	0.613

window in one pass across lags, keeping every lag's summation order, so distances stay bitwise identical; the exact lists equalled the measured build's for 100 of 100 queries at each gallery size [41]. In-database exact search became 1.6–2.7 times faster at p95. With eight workers its p95 was 277, 575 and 1,551 ms on dev galleries of 0.23, 0.50 and 1.03 M windows, against 55, 68 and 238 ms for probe search: probes stayed 5.1–8.5 times faster, with every bootstrap interval above 1.

Against a screened exact path. Most windows cannot be among the nearest, so the product now screens them first [42]. A parallel aggregate over the envelope rows scores every window in float32 against the shifted query and bounds the error rigorously (rounding of the dot products and of the window's mean and spread); only windows whose lower bound is at most the m-th smallest upper bound are re-scored with the unchanged float64 scorer, and the contract's selection rules follow as before. The lists equalled the measured build's and the unscreened path's, bit for bit, for 100 of 100 queries at each gallery size. In one session with eight workers, the screened path's p95 was 138, 221 and 540 ms on 0.23, 0.50 and 1.03 M windows, against 516, 808 and 1,928 ms for scoring every window (3.6–3.7 times faster; 3.2–3.6 with the default worker count) and 88, 130 and 237 ms for probe search: probes stayed 1.6–2.3 times faster at p95 in every configuration, with every bootstrap interval above 1. At 0.23 M windows the median screened search took 77 ms against 64 ms for probes. The machine was busier than in E6, which slowed both paths, so only these within-session ratios are comparisons.

Against a materialized exact scan. Outside the database, exact search can be faster still [40, 51]. All envelopes are held in memory as a float32 matrix; one matrix product scores all 61 lags, a rigorous rounding bound selects the windows to re-score in float64, and the contract's selection rules are applied in the same order. This scan returned exactly the in-database result for every query. Its p95 was 21, 42 and 80 ms on galleries of 0.23, 0.50 and 1.03 M windows, against 68, 108 and 202 ms for probe search at recall 0.99–0.998; no crossover was measured, and fits place one near 34 M windows. The scan holds a 5 GB copy at one million windows, decoded once and scored with the processor's matrix unit, outside the database and without its filters, row-level security or commit-time visibility. Inside that contract, screening brings the same bound into the database, but per-row work remains (reading each row and computing each window's mean and spread at query time): the screened path was 3.5–5.6 times slower than this scan at p95, and probes are the faster path.

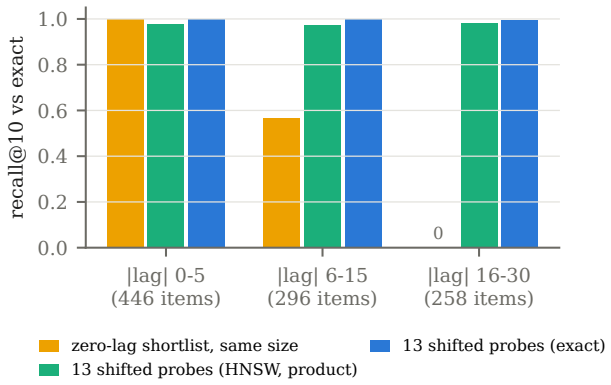


Figure 3: E1, held-out: recall of the exact top ten by the lag at which each item scores best.

Where the loss comes from. Table 3 separates the approximations of Section 3. Each control keeps the 50-row budget per probe and the stored float16 envelopes. On held-out data, exact candidate generation on the sparse grid recovered 0.999 of reference intervals (99.9%); rounding the probes to float16 introduced no further observed loss (0.999). HNSW candidate generation reduced recall to 0.977. Most observed loss therefore arose from approximate candidate selection, with one reference interval already missed by the sparse grid. On dev data the grid lost nothing and HNSW 0.002. A grid at spacing ten lost 0.013–0.014. HNSW with full-precision probes cannot be measured separately, because the `halfvec` index takes float16 queries. Missing probe rows, differences between searched windows and concurrent writes are separate possible sources of loss; they were excluded here by construction and by the checks above. None of this guarantees the same split on other data or with other HNSW settings. The six hardest held-out queries are all high-energy windows of one station, and each reaches 1.0 with exact grid candidates. Figure 3 shows the mechanism: zero-lag candidates find items whose best lag is within ±5 but almost none beyond ±15, whereas probes are flat across lags. Figure 4 shows one case of each kind.

Why spacing five sufficed here. For each of the 1,000 reference items per database, the best of the 13 grid probes kept a median 0.999 of the item's best-lag correlation (5th percentile 0.98, minimum 0.84), and ranked the item at median 2 and 95th percentile 7–8 among all windows in that probe's exact ordering; one held-out item ranked 324th, outside the 50-row budget, which is the single PX-13 miss. On 3,000 dev envelopes the envelope's own biased autocorrelation is 0.973 at an offset of two values and 0.895 at five [39], consistent with the smoothing applied before decimation (about five envelope values). This explains why spacing five works on these data. Recall is decided by the rank inside a probe, so for a new signal the spacing is a measurement, not a consequence of smoothness alone.

Where the time goes. An instrumented copy of the probe function on dev data [31] spent 65% of its time in the 13 index statements (1.9 ms each, p50) and the rest in re-ranking, reading the source interval and routing; instrumentation added about 13 ms to

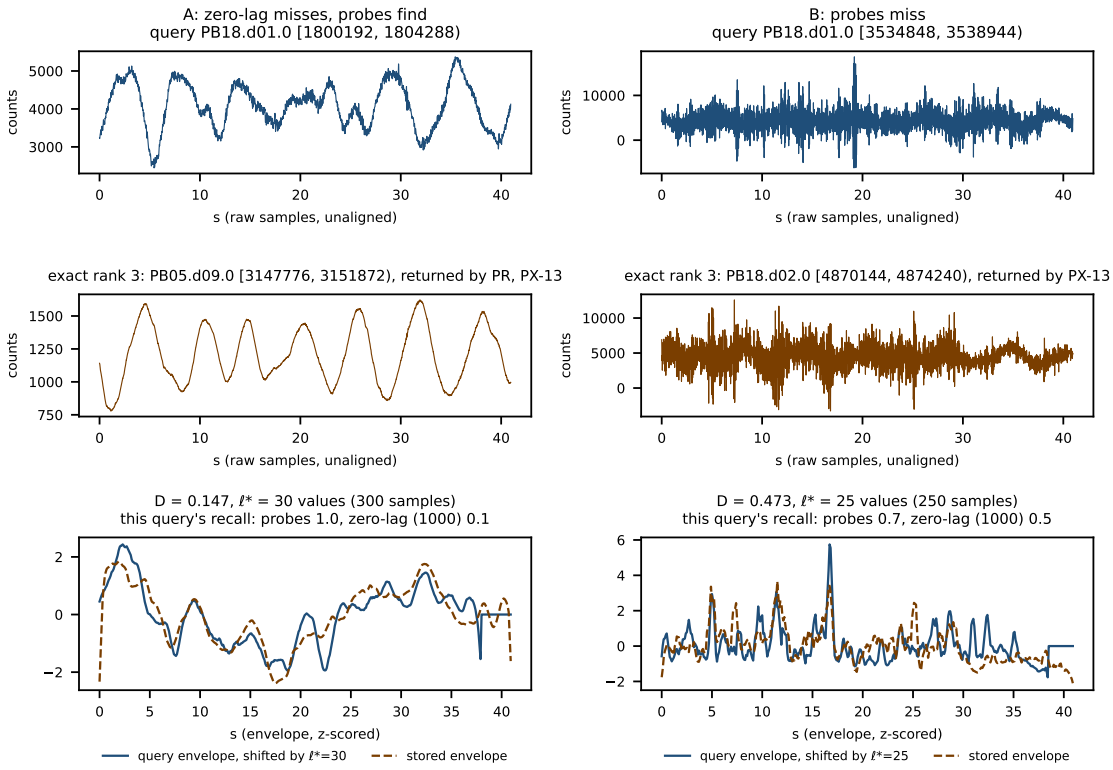


Figure 4: Two held-out cases, selected by a rule fixed before the figure was made: (A) the query with the largest recall gain of probes over a zero-lag shortlist of 1000, and the reference interval with the largest best lag among those the probes find and the shortlist misses; (B) the query with the lowest probe recall, and its best-ranked missed interval, which exact grid candidates (PX-13) do find. They illustrate, and do not replace, the distribution in Table 2; no domain label is implied. Top rows: raw samples in counts, unaligned. Bottom: z-scored envelopes as the scorer uses them, the query shifted by the best lag ℓ^* with zero padding; the distance D is the product scorer's and was checked against $1 - \max_\ell c(\ell)/n$.

the product's 29 ms, mostly by materializing the re-rank, so the index statements account for most of the product's time. The eligibility predicate cost 0.03 ms per statement. Issued on its own, one index statement took 0.38 ms to plan and 0.82 ms to execute (`EXPLAIN ANALYZE`); inside the function each probe statement took about 1.9 ms. Planning inside the function was not measured, so the remaining 0.7 ms per probe is unresolved; planning of the dynamically issued statement, building the probe vector and collecting the result are candidates. ANN traversal, heap access and visibility checks happen inside one index-scan node and were not separated. Every probe returned 50 rows; the union held 335 windows (median), one was excluded as the source, none was suppressed.

Table 4: SS3 (earlier build, shared machine): filtered search on held-out data, mean recall@10 (product: also minimum) against exact search, 100 queries per level. Post-f.: post-filtered probes; F-aware: filter-aware probes; Routed: product function (*routed to exact).

Eligible	Exact p95	Post-f.	F-aware	Routed mean/min	Routed p95
1.8% (4,217)	68 ms	0.515	0.998	**1.0**/1.0*	85 ms
7.2% (16,868)	329 ms	0.848	0.997	**0.997**/0.9	207 ms
25% (58.6 k)	290 ms	0.972	0.986	**0.986**/0.6	98 ms
29% (67.5 k)	390 ms	0.976	0.994	**0.994**/0.8	88 ms
100% (234.6 k)	~1.2 s	0.976	0.977	**0.977**/0.7	69 ms

8 FILTERS AND ROUTING

Investigations usually restrict the gallery to one sensor or period. The first probe implementation applied eligibility after probing. A filtered-search study compared three strategies at five filter levels per query [12]: exact search, post-filtered probes and filter-aware probes; a router threshold was fixed on dev data before the held-out run.

Post-filtering lost up to half the top ten (Table 4). Filter-aware probes restored recall but were slow at the smallest level (p95

724 ms), where the planner abandoned the index and exact search took 68 ms. The router sends that level to exact search. No returned row violated its filter, and with row-level security restricting a role to one station, every strategy returned only that station and rejected a hidden recording as query source. The router missed its pre-registered latency bound once, at the unfiltered held-out level, by 0.8 ms. Low per-query minima at 25% and 100% coincide with the high-energy queries of Section 7, not with routing errors.

Correlated filters. A filter can keep the same number of windows but more or fewer of a query's neighbours, which changes how far an index scan must travel to find eligible rows [34]. A diagnostic chose, per dev query, the day filter (about 7% of windows) or station filter (25%) that held the most, a random, or the fewest of the query's unfiltered top hundred; the choice is synthetic [31]. Recall was 0.997–1.0 in every case. Latency was not: with day filters, probe p95 rose from 121 ms (positive) to 178 (neutral) and 227 ms (negative), against 210 ms for exact search on the now quiet machine; with station filters, probes stayed at 58–63 ms against 264 ms. Routing by the number of eligible windows therefore preserved recall in this workload, but its latency crossover depends on how the filter relates to the query, and the threshold, set on a shared machine, sends 7%-filters to probes where exact search is as fast.

9 STORAGE, RECALL AND LATENCY

Probe search costs space. In E1, raw chunks took 371 MB (dev) and 514 MB (held-out), envelopes 223 MB (read by both paths for scoring), the probe table 223 MB and its HNSW index 275 MB: the probe structures added 1.34× and 0.97× the raw data. Rebuilding the index took 39–41 s with 4 GB of maintenance memory. Smaller storage and faster search are separate goals: float16 probes did not change recall, and storing fewer bytes need not make search faster. How large envelopes are relative to raw data depends on stride and on how well the raw data compress (supplement).

Float16 storage is not the bottleneck: on 2022 windows that no earlier study had read, float16 envelopes kept recall@10 at 0.9999 against envelopes recomputed from raw samples [6]. The two stored copies hold the same float16 values in 98.3% of positions. A study of keeping only one [11] found that a single `halfvec` table was exact on both databases (identical distances for 9.4 M pairs) and saved about 250 MB per 235,000 windows, but made exact scans 23% slower; an HNSW index on an expression over the envelope table kept exact search unchanged but raised probe p95 by 27–32%. Both missed their pre-registered latency bounds. The trade-off is three-way: about 1–1.3× raw storage buys a p95 latency 5–13 times lower than in-database exact search at recall 0.977–0.998; saving about half of that storage costs 20–30% latency on one path.

Maintenance during ingestion. Building vector rows at commit makes new windows searchable immediately; the price is HNSW maintenance on every append. In a replay of one day of four channels, appends ran at 53% of raw-only throughput with every window searchable at its commit [10]. Under sustained load (409,600 samples/s with concurrent searches on 938,768 windows), write-ahead logging was 18 times the database growth with PostgreSQL's defaults and 2.9 times with four settings changed together; a planned two-hour run met its target for 111 minutes and stopped after an unexplained pause [13]. The supplement reports these runs.

10 RELATED WORK AND WHAT IS NEW

We looked for the strongest overlapping prior work, reading primary sources; Table 5 records for each work whether we read the full text (F), parts (P) or only the abstract (A). The full matrix, with quotations and what we searched for, is in the repository [5]. We use *shift* for a displacement along the time axis. Invariance to a vertical offset (adding a constant), as in normalization-based methods [29], and shift-invariant kernels that average over all circular shifts [19] are different notions.

What is known. Three ideas we use are established. (i) *Shifting the query instead of the data.* Rafiei [14] evaluates similarity under a set of transformations, explicitly including time shifts, against one index of untransformed sequences, and processes all transformations in one index traversal; Keogh et al. [15] expand a query into all its rotations, keep the database stored once, support rotation-limited queries by dropping rows of that matrix, and merge neighbouring rotations into a few envelopes, an exact counterpart of our sparse grid. Foulds et al. [17] search under cyclic permutations of the query. All three are exact, use Euclidean-type distances, and treat shifts as circular. (ii) *Several vector queries against one approximate index, union, exact re-ranking.* ColBERT's end-to-end retrieval issues one ANN search per query vector, takes the union of document identifiers and re-ranks it exhaustively [24]. (iii) *ANN candidates for cross-correlation.* In seismology, Tibi et al. [22] project waveforms with kernel PCA so that correlation maps to Euclidean distance, search a KD-tree forest and evaluate exact correlation on the returned set (25× faster than linear search at recall above 80%); Gonzales et al. [23] issue several internal queries to compensate for projection error. FAST obtains shift tolerance from densely overlapping data windows [2, 21], and GRAIL learns embeddings of a shift-invariant correlation kernel [19]. The identity between a lag term and an inner product with a shifted signal is textbook; the biased statistic itself appears as NCC_b in the k-Shape work [18]. On the database side, choosing brute force over an index when a filter is selective is standard in vector systems (AnalyticDB-V, Milvus, ACORN, and the documented thresholds of Weaviate, Qdrant, Vespa and Lucene) [25–27]; VBase and pgvector's iterative scans provide filtered index scans inside PostgreSQL [1, 32]; SingleStore-V makes vector search visible at commit by indexing immutable segments and scanning fresh rows exactly [28]; and KV-match and ULISSE return exact subsequence matches as offsets in stored series [29, 30].

Vector search inside PostgreSQL. VBASE exposes an index as an iterator that the executor can resume until enough rows pass the filters, and stops when the traversal moves away from the query [32]; its guarantee is equivalence to a top-K-then-filter plan with the best K, which needs filters that commute with sorting. Our score is a maximum over lags of which only the grid is probed, and greedy overlap suppression does not commute with sorting, so that argument does not carry over; pgvector's iterative scan already resumes within each probe statement, and E1 and its diagnostics found no query whose fixed budget ran out. PostgreSQL-V

Table 5: Closest prior work. Shift: which displacements along the time axis the similarity tolerates. Side: where the shift is applied. Cands.: whether the candidate set is exact for the stated similarity. Read: F full text, P parts, A abstract. [a]exact for the magnitude feature, which itself approximates the rotation distance.

Work	Similarity	Shift	Side	Index	Cands.	Copies, refinement	Read
Rafiei 1999 [14]	Euclidean on normal form, truncated DFT	set of transformations incl. time shifts (circular in DFT)	transformations applied to index entries	R-tree on DFT coefficients, one traversal for all	exact	one copy; false positives removed exactly	F
Keogh et al. 2006/09 [15]	Euclidean (also DTW)	all circular rotations or a limited range	query: matrix of rotations, merged into wedges	lower bounds; VP-tree	exact	one copy; exact refinement	F
Vlachos et al. 2005 [16]	Euclidean on Fourier magnitudes	all circular	data: invariant feature	VP-tree variant	exact[a]	one compressed signature	P
Foulds et al. 2015 [17]	distance over cyclic permutations of the query	all cyclic	query	none; parallel DFT-based search	exact	—	A
k-Shape, TODS 2017 [8, 18]	normalized CC incl. biased variant	all linear shifts (FFT)	pairwise	none	exact	—	P
GRAIL 2019 [19]	kernel over all circular shifts of NCC	all circular	data: learned embedding	any (Euclidean/IP)	approx.	one embedding per series	P
Qiu et al. 2018 [20]	Pearson at zero lag	none	data: learned embedding	Euclidean index	approx.	one embedding	P
MASS, UCR suite [3, 9]	z-normalized ED per subsequence	every offset	scan	none	exact	—	A
FAST, Rong et al. [2, 21]	Jaccard on binary fingerprints	data windows every 1 s	data: dense windows	MinHash LSH	approx.	one fingerprint per overlapping window	F
Tibi et al. 2017 [22, 23]	max-over-lag CC of waveforms	bounded lags	data: kernel-PCA embedding of CC	KD-tree forest	approx.	projected vectors; exact CC on candidates	F
ColBERT 2020 [24]	sum of max inner products	—	query: one vector per token	one ANN index, one search per vector	approx.	union of candidates re-ranked exactly	F
This paper	biased, zero-padded CC	linear, $\|\ell\| \le L$	query: sparse grid of shifted copies	unmodified IP HNSW, one search per probe	approx.	one vector copy + index; union re-ranked with D	—

moves index contents out of PostgreSQL pages and indexes recent rows asynchronously, giving up the WAL-logged, commit-time index visibility that our contract uses [33]. A study of filtered vector search in a PostgreSQL-compatible engine shows that correlation between filter and query neighbourhood, not only selectivity, governs the cost of iterative scans [34]; D2 observed the same effect on latency, not on recall. MorletDB uses no PostgreSQL hooks and no copied internals, the patterns that a survey of extensibility associates with incompatibilities; its dependencies are PostgreSQL 17 and pgvector 0.8 or later [35].

What is new here. Query-side shifting, multi-query candidate generation and selectivity routing are each known. To our knowledge, no prior work combines them for a *bounded, zero-padded, linear* lag scorer on an *unmodified, approximate inner-product index*. Equation 2 lets a bounded-lag score use any inner-product ANN index without training an embedding (unlike Tibi et al. and GRAIL), without replicating windows per shift (unlike FAST) and without lower bounds (unlike Rafiei and Keogh et al.); in exchange, candidate selection is approximate, and we measure its loss. The diagnosis that zero-lag candidates fail through their metric rather than through approximate search, and the ablation that attributes the remaining loss, are to our knowledge new as measurements. On the systems side, MorletDB keeps intervals, filters and row-level security identical on the exact and approximate paths, and makes built windows searchable at the commit that builds them.

The single-traversal alternative. Rafiei and Keogh et al. show that processing all transformations in one exact traversal with lower bounds can beat one search per transformation. Before building probe search we examined the two lag-uniform bounds closest to this design on our scorer [38]: a Keogh-style envelope bound (one wedge over all lags $|\ell| \le L$) and a spectral bound, on 1,826 windows of 2022 data. This was a feasibility study: even under idealized assumptions (the true tenth-best distance as threshold, bounds computed at no cost) their combination pruned 63% of candidates, because it overestimated the best correlation by a median factor of 1.83. Inserted into the measured per-candidate costs of the exact scan of that time, this gives a modelled speed-up of about 1.9× on that gallery and 2.3× in the limit of large galleries. These are model values for one construction, not measurements; finer wedges or other bounds could do better. They explain why we pursued approximate candidates rather than exact pruning.

11 LIMITATIONS

Scope of the claims. We do not claim a new similarity measure or a new index, and probe search has no recall guarantee: its recall is measured, not bounded. Recall is fidelity to the scorer, not relevance to a domain question.

One signal domain. All retrieval results are for seismic envelopes from one network, and our empirical claims are limited to these data. The method needs only a bounded-lag inner-product scorer, but how many probes and candidates suffice depends on the representation's smoothness and on where best lags fall relative to the grid; we have not measured this on vibration, acoustic or physiological data. Condition monitoring motivated the work and is a target application; we have no evidence that probe search, or envelope similarity in general, is diagnostically relevant for machine condition.

Earlier implementation states. E1 measured the main comparison on one build. The filtered-search table (SS3), the single-copy study (SS1), the zero-lag scaling study (R2, R2-D) and the load runs were measured on earlier states and on a machine shared with another job; ratios across those studies mix states.

Speed relative to exact search. The speed-up is against in-database exact search with results identical to the product's. Held-out speed-ups (Table 2) are against the measured build; against the current screened exact path, probes were 1.6–2.3× faster at p95 on dev galleries up to 1.03 M windows. A materialized exact scan outside the database was 2.5–3.3× faster than probe search at p95 up to 1.03 M windows, without the contract.

Measurement conditions. One laptop-class machine, warm caches, one HNSW configuration; only EX2 ran on other hardware, and it measured ingestion, reads and search speed, not retrieval quality. The load tests ran 15 and 111 minutes; the planned two-hour run stopped after an unexplained pause, and we make no claim about longer operation, archives beyond about one million windows, cold caches or several tenants under load.

What recall means. Recall@10 measures fidelity to the scorer D. Whether the bounded-lag top ten are the intervals an analyst wants is a separate question. A proxy task gives first evidence for the scorer: with a query window containing a catalogued earthquake's predicted P arrival, the exact bounded-lag top ten contained such windows in 0.297 of positions against 0.213 for zero lag on development data (465 queries from 171 EMSC events; difference +0.084, 95% interval $[0.070, 0.098]$ by event bootstrap), and 0.139 against 0.060 on held-out stations and year (116 queries from 92 events; +0.078, $[0.054, 0.105]$), against a chance level below 0.005; both frozen decision rules read "bounded lag better" [43]. Catalogue membership is a proxy for relevance, the absolute precision is modest, the comparison used exact search only, and we have no task evaluation of probe search; given its recall, we expect but did not measure a similar gap.

Approximation without bounds. Probe search can miss a window that the exact path returns (per query, down to 7 of 10 on held-out data); the budget is fixed and is not extended when exclusion or suppression removes candidates. The idealized lower-bound study of Section 10 covered one construction; finer bounds were not tested.

Routing. The router's threshold counts eligible windows only. The correlated-filter diagnostic showed that its latency crossover depends on how the filter relates to the query and on machine load; we did not recalibrate it.

Grid choice. The 13-probe grid was chosen on dev data and explained post hoc by envelope smoothness; it was confirmed on held-out data, for this representation only.

When to use it. Exact search for small galleries (a few thousand eligible windows), for selective filters that exclude the query's neighbourhood, and whenever every true neighbour must be found. With screening, exact search took 0.14–0.54 s at p95 on dev galleries of 0.23–1.03 M windows, so it is an interactive option up to about a million windows when every true neighbour matters. Probe search for unfiltered or weakly filtered galleries of a quarter of a million windows and more, where it was 1.6–2.3 times faster at p95 than the screened exact path at recall 0.99–0.998 (Section 7), and for larger archives, since exact search reads every eligible window. When a separate in-memory copy without filters or security is acceptable, a materialized exact scan is faster still. The crossover points measured here hold for these data and this machine.

12 CONCLUSION

Bounded-lag waveform search can use a standard inner-product index if the index is asked the right question. A zero-lag shortlist asks a different one and loses a quarter of the scorer's top ten, whether it is computed exactly or approximately. Because each lag term is an inner product with a shifted query, thirteen shifted probes into one unmodified HNSW index, re-ranked with the unchanged scorer, recovered 97.7% of the top ten on held-out seismic stations and year, and stayed 5–8.5 times faster at p95 than an in-database exact path made 1.6–2.7 times faster with identical results. Exact candidates on the sparse grid recovered 99.9%; float16 probes added no observed loss; approximate candidate selection accounts for the rest. The method is simple: one index, one stored copy of each window, no training and no lower bounds. Inside PostgreSQL it keeps one contract on both paths: hits are sample intervals of losslessly stored recordings, filters and row-level security apply, and windows become searchable at the commit that builds them. Its costs are measured: probe structures of 1.0–1.3× the raw data, exact search for small and selective galleries, and write-ahead logging from index maintenance (18 times the database growth with default settings, 2.9 times at a tuned operating point). On a catalogued-earthquake proxy task, the bounded-lag scorer found more windows with earthquakes than zero-lag similarity on development and held-out data. A second signal domain is the next step for claims beyond seismic data.

USE OF AI ASSISTANCE AND RESPONSIBILITY

Code, experiment harnesses, analyses and text of this report were developed with Claude (Anthropic), an AI assistant, working under the author's direction. Where a study had a protocol, it was committed before measurement (Section 5 and the supplement); the author reviewed the results and is responsible for the content and for any errors.

LICENCE

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A CLAIMS AND EVIDENCE

Each claim with the study that supports it, the measurement kind (E: offline experiment control, P: product functions) and data (D: dev, C: held-out), and its main restriction. Protocols, results and commands: [5]; E1 and diagnostics: [31].

Claim	Study	Restriction
zero-lag top 1000 + re-rank: 0.743 (exact), 0.744 (HNSW)	R2-D; E, D	earlier build; two budgets
HNSW and exact shortlists share 98–100% of candidates	R2-D; E, D	largest gallery
zero-lag HNSW 1000 from probe index: 0.744 / 0.658	E1; P, D / C	one HNSW configuration
probes 0.977 (min 0.7; 5/100 <0.9), p95 57 ms	E1; P, C	seismic; unfiltered; quiet machine
fastest in-database exact p95 740 ms (8 workers allowed)	E1; P, C	PG 17 defaults, warm cache
materialized exact scan p95 21/42/80 ms vs probes 68/108/202 ms at 0.23/0.50/1.03 M windows	E3, E4; E, D	outside the database; no filters, RLS
faster exact path (identical lists) p95 277/575/1551 ms vs probes 55/68/238 ms	E6; P, D	8 workers; float64 identical
screened exact (identical lists) p95 138/221/540 ms vs every window 516/808/1928 ms vs probes 88/130/237 ms	E7; P, D	8 workers; one session, busier machine
probes 0.998, p95 53 vs 783 ms	E1; P, D	development data
exact grid candidates 0.999 (1 of 1,000 missed); float16 probes no further loss	E1; E, C	offline controls
HNSW candidates: 0.999 → 0.977 (held-out)	E1; P vs E	this configuration only
random 0.998, high-energy 0.956 (held-out)	E1; P, C	energy is a query rule, not relevance
83/12/4/1 held-out queries at 1.0/0.9/0.8/0.7	E1; P, C	3 of 23 misses were rank 1
same complete universe on both paths	E1 check	static databases
7-lag grid loses 0.013–0.014	E1; E, D+C	offline control
grid probe keeps ≥0.98 correlation (p5), rank ≤8 (p95)	E1; E, D+C	explains, does not guarantee
no short results; identical lists on the same index	E1; P, D+C	rebuilt index may differ

Claim	Study	Restriction
index statements ≈65% of probe time; 0.7 ms per probe unresolved	D1; P, D	instrumented copy
correlated filters: recall 0.997–1.0, p95 121–227 ms	D2; P, D	synthetic filter choice
post-filter 0.515 at 1.8% eligible; routed 1.0	SS3; E/P, C	earlier build
routed recall 0.977–1.0, no filter or RLS violations	SS3; P, D+C	earlier build
probe structures 0.97–1.34× raw; rebuild 39–41 s	E1; P	depends on raw compressibility
one stored copy: −250 MB, +23% or +27–32% p95	SS1; E, D+C	earlier build
lag-uniform bounds prune ≤63% (idealized)	pruning; E	one construction; model
searchable at commit; 919k samples/s	BL1-S; E, D	one day, four channels
409.6k samples/s: 15 min (defaults), 111 min (tuned)	C2, C4; P	2-h run not completed
WAL/growth 18× (defaults), 2.9× (second point)	C2, C2p; P	four settings and load differed
companion-table RLS: exact +5%	Stage 1; P	one tenant layout
calibrated floats: codec 0.62–0.76× ClickHouse, TileDB	B2; E	six inputs; ClickHouse 26.5
smallest on 4 of 6 waveforms; Pcodec 5–8% smaller on 2; ALP 2.6–7.9× larger on calibrated	B4; E	default settings; one machine
grid detected without metadata: 0.98–1.00× declared-quantum size	G1; E	six inputs
sample data 33% / 87% smaller than QuestDB Parquet zstd 9 / native	B3; P	one dataset; needs a value grid
QuestDB ingests 1.2–1.6× faster (one connection); MorletDB 1.2–2.0× (parallel)	B3.2; P	one machine; 8-CPU containers
search features at ingest: 3.9× raw ingest time (was 9.6×)	B3.3; P	one connection, one dataset
seven-channel acquisition reads 32–39% faster than QuestDB (p50)	B3.3; P	one dataset
summaries: statistics 58× faster (p95, 60 s at 64 kHz), bitwise identical; ingest −4.5%, storage +1.3%	CS1; P	gain grows with interval length
full workflow at 2 CPUs, 512 MB, no slowdown	EX1; P	container limits on desktop cores
Raspberry Pi 3B+: 78k / 258k samples/s, search lists identical	EX2; P	one device, no fan; not long-term
bounded lag P@10 0.297 vs 0.213 (dev), 0.139 vs 0.060 (held-out)	E5, E5-H; E, D / C	catalogue proxy; exact search